

underlying code of an application from the interface code itself. For example, while creating an accounting software, the programmers need to work closely with the interface designers to make sure that the buttons and scrollbars and so on, that the interface designers use to build the face of the program; are coded properly. Thus a programmer needs to work closely with artists and interface designers to ensure that parties involved are on the same page. This is an obvious bottleneck which is alleviated by Vista by offering different code paths for interface and backend. These paths are exposed via a markup language similar to HTML and leveraging XML technology—XAML (Extensible Application Markup Language) allows designers to create an application interface using XML over which the programmers can then either separately plug their code in, or embed the code into the XAML document itself.

2.2.4 All Those Left Behind

There are several implications going this route: on a software level, all future software will need to be .NET applications to take full advantage of Vista. This is since all APIs are written in .NET Framework 3.0; this is the first step to migrate the entire Windows ecosystem towards managed code. On a hardware level, this new design calls for an up-to-date video system for a PC. The video card, whether dedicated or onboard, would require at least 64 MB of RAM in order to serve the graphical whiz-bang Vista brings.

Apart from the memory requirement, the cards also need to support DirectX 9.0 and above—essentially, cards which lend pixel-shading support to Direct3D. A pixel shader is used to apply textures and special effects to a screen element; support for a pixel shader is needed in order to pull off effects such as transparency and shadow for a window. Apart from this direct hardware support, Vista also requires a new graphics driver in order to work as required. Under Vista a video card's driver can be one of two models: the old XP-style driver, termed XPDM (XP Driver Model) and a driver model made especially for Windows Vista called WDDM (Windows Vista driver model). A video card offering a WDDM driv-